Proposed next experiment — only one untested code deviation remains

What 6+ weeks of audit found

MATCHES source code:

LR schedule · D head
AMP precision · L2-normalize
BCE losses

Engineering tradeoffs:

HuBERT freeze (DDP deadlock fix)
→ single-GPU run shows 3.51%
ruling this out as gap source

REAL untested deviation:

λ_adv control law
(cap, floor, per-batch reset)

The diff

Source code (paper)

```
every G step:
  λ_adv = 0.25        # reset!
  if ratio > 1.5:  λ = min(λ·1.1, 0.01)
  if ratio < 0.5:  λ = max(λ·0.9, 0.0001)

Effective range:  {0.01, 0.225, 0.25}
```

Our code (current)

```
init: λ_adv = 0.25  (NO per-batch reset)
  if ratio > 1.5:  λ = min(λ·1.1, 0.5)
  if ratio < 0.5:  λ = max(λ·0.9, 0.01)

Effective range:  [0.01, 0.5] with memory
Drift: G crushed → λ decays to floor
```

Proposed experiment

exp/fix-lambda-adv-source-control-law-ddp
(branched from exp/innovation-specaug-ddp, current best 3.27%)

train.py — 3 changes:
(a) every G step: self.lambda_adv = 0.25 # reset
(b) cap: 0.5 → 0.01
(c) floor: 0.01 → 0.0001

Expected outcomes (both informative):

If EER drops 3.27 → ~3.10:
λ_adv control law IS the gap.
Lock as new default. Write into paper.

If EER stays ~3.27:
Gap is seed noise. Run multi-seed next.

Cost: 1 branch · ~8h on 4-GPU DDP · single decisive experiment